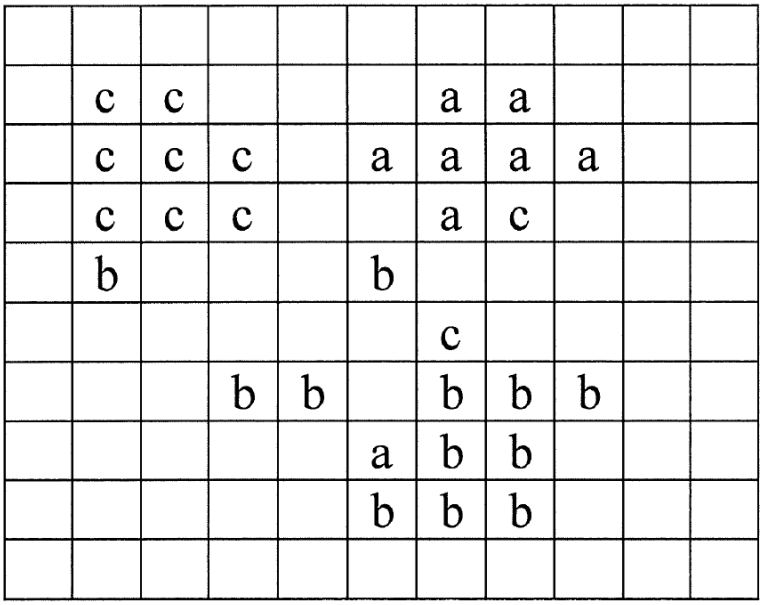
Tutorial 2

1. Find the Eigen values and Eigen vectors for the matrix

$$\begin{pmatrix} 3 & 0 & 1 \\ -4 & 1 & 2 \\ -6 & 0 & -2 \end{pmatrix}$$

2. Apply 9-NNR to the following data using chessboard distance measure.



3. One student was asked to determine the face make-up of a person. Let us assume that the 2 classes have equal probability. Three features were extracted from face images with large number of samples. The following shows a collection of 8 people.

	Face contrast	Lip hue index	Eye lid contrast
Wear make-up	2500	0.71	11200
	2670	0.78	11100
	2300	0.87	11200
	2450	0.90	23000
No make-up	1980	0.45	9004
	1833	0.55	8500

	2111	0.63	9100
	2023	0.73	5500

Face contrast – take the mean of the face first, and then compute the variance

Lip hue index – turn the RGB image into HSV, where S stands for saturation. It is a measure of colourness, ranging from 0 to 1. Where 0 means no colour (or B/W) and 1 means full colour.

Eye lid contrast – detect the eye region and find contrast.

Train the classifier and determine the make-up with the input of (2000, 0.72, 9500).